

Exercise 10

Date:

Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using javascript

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

First, create an HTML file to hold your canvas for the chart and include Chart.js.

html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Inventory Management Visualization</title>
```

```
  <style>
```

```
    body {
```

```
      font-family: Arial, sans-serif;
```

```
      text-align: center;
```

```
      margin: 50px;
```

```
    }
```

```
    canvas {
```

```
      margin: 20px auto;
```

```

    }
  </style>
</head>
<body>
  <h1>Inventory Management System</h1>
  <canvas id="pieChart" width="400" height="400"></canvas>
  <canvas id="barChart" width="400" height="400"></canvas>
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <script src="script.js"></script>
</body>
</html>

```

Step 2: Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic.

```

javascript
// script.js

// Data for the inventory
const inventoryData = {
  labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
  datasets: [
    {
      label: 'Items in Stock',
      data: [200, 150, 100, 80, 50],
      backgroundColor: [
        '#FF6384',
        '#36A2EB',
        '#FFCE56',

```

```
        '#4BC0C0',  
        '#9966FF'  
    ],  
    }  
]  
};
```

// Creating the Pie Chart

```
const ctxPie = document.getElementById('pieChart').getContext('2d');  
const pieChart = new Chart(ctxPie, {  
    type: 'pie',  
    data: inventoryData,  
    options: {  
        responsive: true,  
        title: {  
            display: true,  
            text: 'Inventory Distribution'  
        }  
    }  
});
```

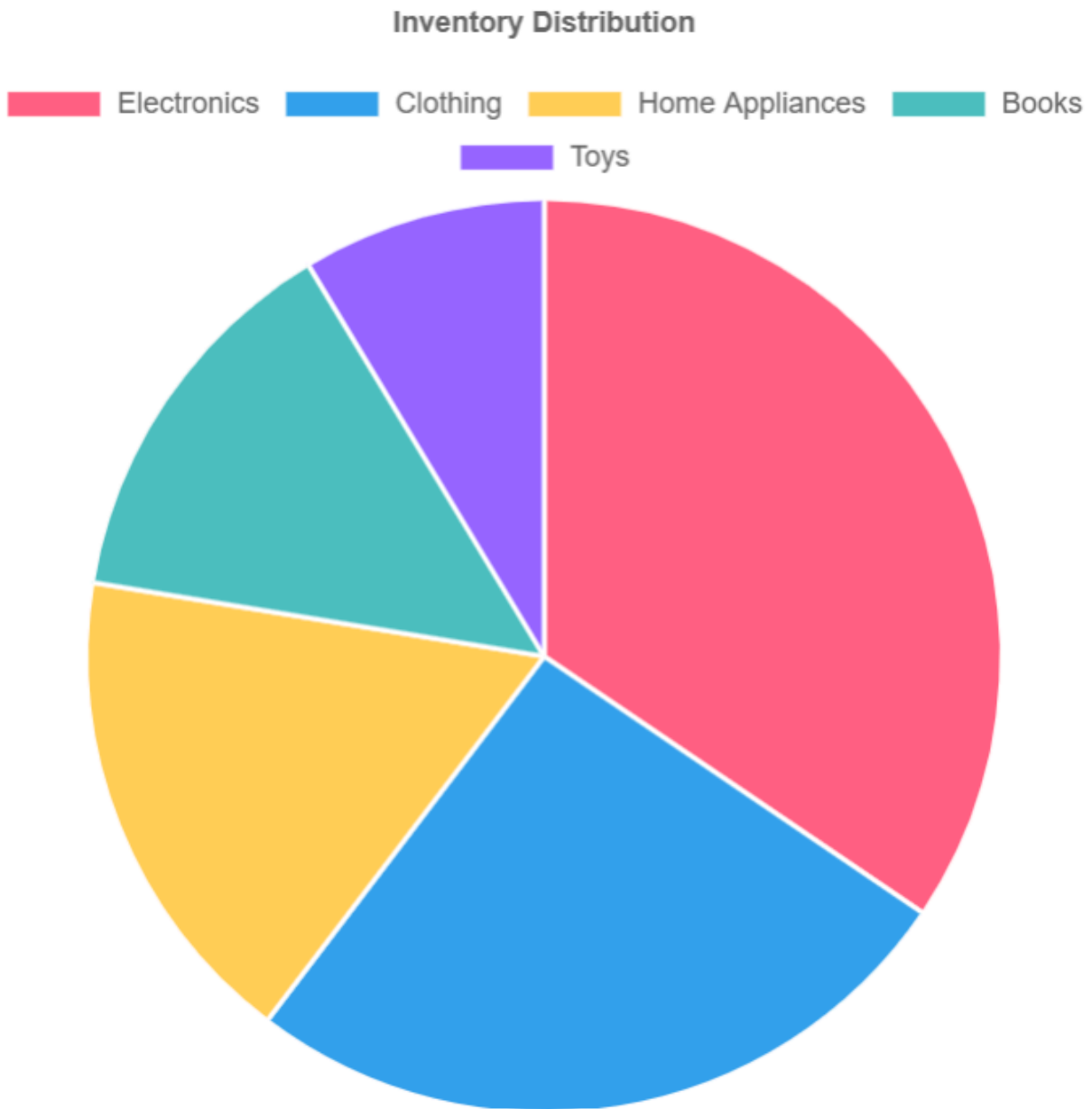
// Creating the Bar Chart

```
const ctxBar = document.getElementById('barChart').getContext('2d');  
const barChart = new Chart(ctxBar, {  
    type: 'bar',  
    data: inventoryData,  
    options: {  
        responsive: true,  
        title: {
```

```
    display: true,  
    text: 'Items in Stock by Category'  
  },  
  scales: {  
    yAxes: [{  
      ticks: {  
        beginAtZero: true  
      }  
    }]  
  }  
}  
});
```

OUTPUT:

Inventory Management System



RESULT:

Hence the Inventory management system with pie charts is implemented and executed.